

Mark Gascoyne

Polyglot Full Stack Software Engineer

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Summary

Polyglot Engineer with 16+ years building production systems across the full stack: Golang, Python, TypeScript, Kubernetes, and cloud-native infrastructure. I architect distributed systems, build Kubernetes operators, and establish GitOps practices. I am co-founder and CTO of Predictive Cloud, where I built and run PredBat, a multi-tenant Kubernetes platform for home battery optimisation, spanning custom ESP32 gateway hardware and firmware through to iOS and Android apps. Alongside that I contract: most recently the pricing and quote-binding services behind Tempcover's next-generation quoting platform, and a Kubernetes sandbox substrate with MCP tooling for AI agents at Testkube. Before contracting I led a team of 4 engineers at Vitrifi, delivering eventually-consistent datastores for multi-region network synchronisation and 5+ production-grade Kubernetes operators. I thrive on complex technical challenges and can work at any level of the stack.

Skills

DevOps & Orchestration

Kubernetes | Helm | Flux | ArgoCD | Skaffold | Kustomize | Docker | Linux | Terraform | Argo | K3S | k3d

Programming Languages

Golang | TypeScript | .Net | CSharp | VB.Net | Scala | Node.js | Java | Bash | Python

Tools & Frameworks

GitHub | Git | Circle CI | TFS | MVC | Spring | Confluence | GitLab | BitBucket | Jira | Model Context Protocol | GitHub Actions | k6

Web & Frontend Development

Vue.js | React | Swagger | Rest | Signalr | JQuery | Knockout | Backbone.js | Highcharts | Soap | gRPC | GraphQL | OpenAPI | Capacitor

Databases & Messaging

Sql Server | Cassandra | Entity Framework | Redis | Microsoft SSAS | SQL Server Integration Services | Apache Kafka | NATS | Supabase | PostgreSQL | MongoDB | DynamoDB | MQTT

Hardware & Embedded

Z Wave | Modbus | ESP32

Cloud Platforms

Google Cloud | Azure | AWS | Confluent Cloud | Civo | Oracle Cloud Infrastructure | AWS Lambda | Cloudflare

Monitoring & Observability

Prometheus | Grafana | Thanos | Loki | OpenTelemetry

Experience

Predictive Cloud Limited

Remote

Co-founder & CTO

Feb 2025 - Present

Co-founded the company and lead all technical delivery, from the multi-tenant Kubernetes platform and customer-facing applications through to custom gateway hardware and firmware. Set technical direction and own production operations for a live subscription service with a recurring revenue base.

Achievements

- Architected and operate a multi-tenant Kubernetes platform on Hetzner, with per-customer namespace isolation, Flux GitOps, templated per-customer manifests and SOPS/age encrypted secrets, serving 280+ paying subscribers in production.
- Designed the PredBat Gateway, a custom ESP32-S3 USB dongle speaking Modbus RTU directly to inverters, taking it from hardware specification through schematic capture and contract fabrication to firmware, an over-the-air update pipeline and Bluetooth LE provisioning.
- Built the device telemetry pipeline, ingesting samples over MQTT into VictoriaMetrics with Grafana dashboards, alerting and public status pages.
- Integrated UK energy suppliers and inverter vendors including Octopus, EDF and E.ON, alongside GivEnergy, Fox ESS, Solis, SolaX, Sunsynk, Deye and Ohme, covering OAuth flows, token refresh and reverse-engineering undocumented APIs where no public contract

existed.

- Delivered the customer-facing product as a React and TypeScript dashboard with real-time WebSocket telemetry, Supabase authentication, subscription billing, internationalisation, and iOS and Android applications via Capacitor with end-to-end Playwright coverage.
- Reverse-engineered inverter firmware for interoperability, building a static analysis toolchain that recovers Modbus register maps and scaling factors with per-fact confidence scoring.
- Secured cluster access behind a Cloudflare Zero Trust tunnel with K3s OIDC validating access tokens, gating administrative access on GitHub organisation membership and eliminating public API and SSH exposure.

Skills

kubernetes | go | python | typescript | react | terraform | flux | kustomize | supabase | postgresql | mqtt | grafana | esp32 | modbus | capacitor | cloudflare | docker

Tempcover

Remote

Senior Software Engineer (Contract)

May 2026 - Aug 2026

Built out the next-generation quoting platform: a set of Golang services on AWS covering pricing enrichment, quote persistence and binding, and the bridge to the legacy estate, working across service boundaries and coordinating contract changes with neighbouring teams.

Achievements

- Built the pricing computation layer for a new quote enrichment service end to end, covering request builder, rating engine adapter and response mapper, replacing stubbed pricing with live rating calls, then extended scheme coverage from private car to commercial vehicle and courier.
- Designed a profile-based fixture system and an in-process deterministic mock rating server, giving the team fast, reproducible integration tests without depending on a vendor staging environment.
- Added persistence to the quoting API with DynamoDB quote and bound-quote tables (TTL, point-in-time recovery, production deletion protection), a commit endpoint using conditional writes to guarantee single-bind semantics, and quote retrieval by both native and legacy identifiers via a secondary index.
- Built a backfill client from scratch to bridge the next-generation platform to the legacy estate, covering field mapping, create and update calls wired into the create and commit paths, stage-gated rollout, and narrowly typed error sentinels so each failure mode surfaces a distinct API response and log signal.
- Authored OpenAPI 3.0.3 contracts for both services with CI validation of cross-repo reference bundling, led a coordinated schema migration across three services, and extracted a shared underwriters package and service client to replace hardcoded lookups in four more.
- Diagnosed and fixed three production defects in the rating engine request path, covering correlation header propagation, postcode normalisation and timezone handling, shipping the riskiest behind a killswitch, and produced standalone reproduction scripts that proved a hard concurrency ceiling in the vendor environment and isolated a front-end failure that never reached the backend.
- Raised the engineering baseline across the estate with k6 smoke and service tests in PR checks, conventional commit validation, consolidated git hooks running golangci-lint and gofmt, and PII scrubbing from application logs.

Skills

golang | aws | dynamodb | aws-lambda | openapi | k6 | opentelemetry | github-actions | docker | typescript

Testkube (Kubeshop)

Remote

Senior Software Engineer (Contract)

Feb 2026 - May 2026

Worked across the Testkube control plane, agent, and AI copilot, covering Golang backend services, Kubernetes controllers, gRPC APIs, and the local development platform used by the wider engineering team.

Achievements

- map[Built the Runspace agentic execution substrate:a gRPC bridge service plus a Kubernetes agent-sandbox controller that provisions per-session sandbox pods, allowing AI agents to execute commands inside an isolated cluster workload, with JWT bridge authentication, ownership enforcement, and an end-to-end lifecycle smoke test.]
- Extended Testkube's MCP server with tools for execution tag editing, execution ID lookup, and ranged artifact retrieval (line ranges, grep, binary detection), substantially reducing token consumption when AI agents read large test artifacts.
- Designed and delivered AI cost guardrails including organisation-level token caps, per-agent token limits, mid-session enforcement, and per-trigger rate limiting.
- Delivered webhook agent-targeting end-to-end, spanning CRDs, domain models, mappers, and target-selector filtering in the public API.
- Replaced the legacy local development setup with a declarative GitOps workspace built on k3d, Tilt and Kustomize components, adding live-reload Go builds and Vite HMR for the dashboard to shorten the inner loop for the whole team.
- Hardened the core platform with agent key rotation with grace periods, fail-fast handling for parallel workflow steps, log-stream retries on kubelet proxy errors, insecure registry support for image inspection, additional linting, dockerised OpenAPI code generation, and

automated PR review reminders.

Skills

golang | kubernetes | grpc | mcp | kustomize | tilt | k3d | helm | postgresql | mongodb | nats | docker | github-actions |
react | typescript

Vitrifi

Remote

Lead Software Engineer (converted to perm)

Jun 2023 - Dec 2025

Led a team of 4 engineers, designing and implementing scalable software architectures while contributing hands-on to critical software development. Reviewed code submissions to ensure adherence to coding standards and quality. Managed project timelines and collaborated with product managers to align technical and business objectives.

Achievements

- Designed and implemented distributed CRDT-based datastore system for multi-region network state synchronisation using Golang, achieving eventual consistency through conflict-free replication without requiring consensus protocols, with automatic membership tracking across decentralised nodes.
- Architected and developed 5+ production-grade Kubernetes operators using Golang for network service lifecycle management.
- Built GitOps infrastructure using ArgoCD and Kustomize serving as single source of truth for all production deployments, eliminating configuration drift.
- Established platform engineering capabilities enabling development teams to self-service infrastructure through GitOps workflows.
- Pioneered SRE practices including observability, monitoring, and automated incident response procedures.
- Led technical strategy and mentored team members on Kubernetes, Golang, and distributed systems best practices.

Skills

aws | oci | golang | kustomize | argocd | k3s | kubernetes | docker | gitlab | bash | terraform | python

Vitrifi

Remote

GitOps Consultant (Contract)

Dec 2022 - Jun 2023

Implemented GitOps strategies for the organization and contributed to the development of custom tools designed to integrate with GitLab CI. Adapted and extended an existing library of internal tools to fit the unique requirements of the business, enhancing deployment automation and efficiency.

Achievements

- Implemented GitOps across multiple environments, automating deployment pipelines using GitLab CI.
- Developed and customized tooling to streamline GitOps workflows, including writing new tools from scratch to meet specific business needs.
- Enhanced deployment reliability and scalability by integrating custom tools with existing infrastructure.

Skills

aws | golang | kustomize | argocd | flux | k3s | kubernetes | docker | gitlab | bash | terraform

Vitrifi

Remote

Senior Go Developer (Contract - initial engagement)

Jun 2022 - Dec 2022

Developed CI/CD pipelines with Gitlab CI and a CNE Kubernetes operator.

Achievements

- Successfully delivered CI/CD pipelines and CNE Kubernetes operator.

Skills

aws | golang | kustomize | argocd | flux | k3s | kubernetes | docker | gitlab | bash | terraform

Dae.mn/ Kultralab

Remote

Senior Software Engineer

Jun 2021 - Jun 2022

I delivered infrastructure as code and declarative infrastructure maintained using GitOps, and delivered CI/CD pipelines. I refined the backlog, gathered requirements, and developed the supporting services.

Achievements

- Successfully delivered infrastructure as code and CI/CD pipelines.

Skills

aws | postgresql | go lang | skaffold | kustomize | flux | kubernetes | terraform | k3s | react | graphql | docker | bitbucket | jira | confluence | bash | terraform

Ve Global

Remote

Software Consultant

Jul 2020 - Jun 2021

I restructured Ve’s infrastructure and technology to save costs and create a more manageable environment. I introduced and set up local development using Kustomize/Skaffold/Minikube, and rolled out GitOps to improve deployment pipelines. I also trained developers to develop best-in-class microservices.

- Achievements
- Successfully restructured infrastructure and technology.
 - Implemented GitOps to improve deployment pipelines.
 - Coached development teams on microservices best practices and modern engineering workflows.

Skills

azure | kubernetes | skaffold | kustomize | vuejs | kafka | confluent-cloud | go lang | docker | flux | bash

Thermeon Worldwide Ltd

Remote

Software Engineer

Jan 2018 - Jul 2020

I worked closely with both the development and infrastructure teams, which are spread all over the world. I engineered microservices written in Golang and delivered them via the Google Kubernetes Engine. I introduced operations via pull requests to deliver continuous deployment, maintaining stability.

- Achievements
- Undertook the company's first leap into Kubernetes.
 - Worked with all teams to introduce an agile way of working.
 - Developed and delivered a single sign-on solution hosted in Kubernetes and written in Golang.
 - Developed and delivered a PCI-compliant cryptographic encryption microservice in Golang, implementing secure data handling and encryption algorithms for payment card data protection.
 - Introduced GitOps using Weaveworks and various other Kubernetes technologies (cert-manager, external-dns, sealed-secrets).
 - Produced a Horizontal Pod Autoscaler for Kubernetes that uses pod readiness rather than metrics for scaling.
 - Developed a client-facing website for booking rentals.

Skills

github | circle-ci | kubernetes | go lang | helm | flux | vuejs | google-cloud | docker | linux | cassandra | swagger | rest | bash

Utilitywise PLC

Chesterfield, UK

Senior Software Engineer

Apr 2015 - Jan 2018

Delivered control and automation systems for the Wiselife IoT product, enabling smart home device management and energy monitoring.

- Achievements
- Engineered IoT control systems integrating Z-Wave and Modbus protocols for home automation and energy management.
 - Built data ingestion pipelines and analytics services processing smart meter and IoT device data.
 - Developed real-time monitoring dashboards enabling customers to track energy consumption patterns.

Skills

entity-framework | sql-server | z-wave | modbus | cassandra | scala | dotnet | entity-framework | nodejs | docker | linux | tfs | signalR | redis | git | azure | aws

t-mac Technologies Ltd

Chesterfield, UK

Software Engineer

Apr 2012 - Apr 2015

Developed monitoring, controls, and analysis software for the energy sector in a fully agile environment. Delivered bespoke software solutions for major UK energy companies.

Achievements

- Designed and developed customised energy dashboards for major energy providers, enabling real-time consumption monitoring and analytics.
- Built the t-mac energy analytics platform providing energy consumption insights and optimization recommendations for commercial clients.
- Engineered data ingestion and processing services handling smart meter data streams and energy usage analytics.

Skills

csharp | mvc | java | spring | jquery | entity-framework | knockout | backbone | highcharts | ssas

EnServe Group Limited

Leeds, UK

Software Engineer

Sep 2010 - Apr 2012

I worked under the direction of Richard Marr (Senior Lead Developer) on the development and maintenance of Job Track Enterprise (JTE) for H2O Water Services Ltd and Thames Water / Veolia Water. I was fully involved in the maintenance and ongoing development of the JTE system for H2O, including the Smartclient application, website, web services, mobile devices database, and SSIS packages. I was also fully involved in the development of the JTE system for Thames Water in the run-up to live deployment, including all of the above elements plus integration with the IBM Websphere messaging system for near real-time integration with the Insight TMA system. Additionally, I interpreted new requirements for both JTE systems in consultation with internal and external stakeholders.

Achievements

- Developed and maintained the JTE system for H2O and Thames Water.
- Integrated the JTE system with IBM Websphere for near real-time data processing.

Skills

soap | csharp | sql-server | vbdotnet | ssis

Cse-Healthcare Systems Ltd

Eckington, UK

Student Software Engineer

Jun 2008 - Sep 2009

I maintained product specialist’s demo laptop builds of RiO and developed and maintained internal tools used by all areas of CSE-healthcare systems. I worked as part of the live service team investigating software issues, issuing hot fixes and releases, and as part of the testing team, testing software modules, identifying code paths, and creating testing plans. I also designed and created the replacement correspondence reference system, which is used company-wide, and a testing automation framework that automated web pages from the user point of view.

Achievements

- Designed and created a testing automation framework that was flexible, reusable, and well-documented.

Skills

csharp | sql-server | ssis

Education

Sheffield Hallam University

BSc(Hons) Software Engineering

2006-2010

Interests

I possess a diverse array of skills and interests that keep me engaged and constantly learning. Here are just a few of my passions:

Family Adventures: I cherish quality time with my family, especially when it involves taking my son on camping and hiking excursions.

Overlanding Enthusiast: I love exploring off-grid and hard-to-reach places in my 4x4, discovering new terrains and enjoying the thrill of adventure.

DIY Electronics and Home Automation: I’m passionate about creating innovative solutions through DIY electronics and home automation projects, always looking for ways to enhance our living space.

3D Printing and Design: I engage in 3D printing and CAD/CAM design, combining creativity and technology to bring unique ideas to life.